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1- DESCRIPTION

This document describes replacement of the Medcam Array Processor (AP) board, located in the ISE/TPS Chassis. There are currently 3 different varieties of this board in the installed base. Each board has a part number label affixed to the bottom of the foreplane. (See Illustration 1.) Not all the Medcam baseboards support all the available components. Be careful to only use compatible components when replacing the Medcam baseboards or parts on Medcam baseboards. (See the compatibility chart in Table 1.) More compatibility information can also be found in *Service Note 67036, Introduction of New Medcam Memory and Baseboard*, included in Appendix A of this document.

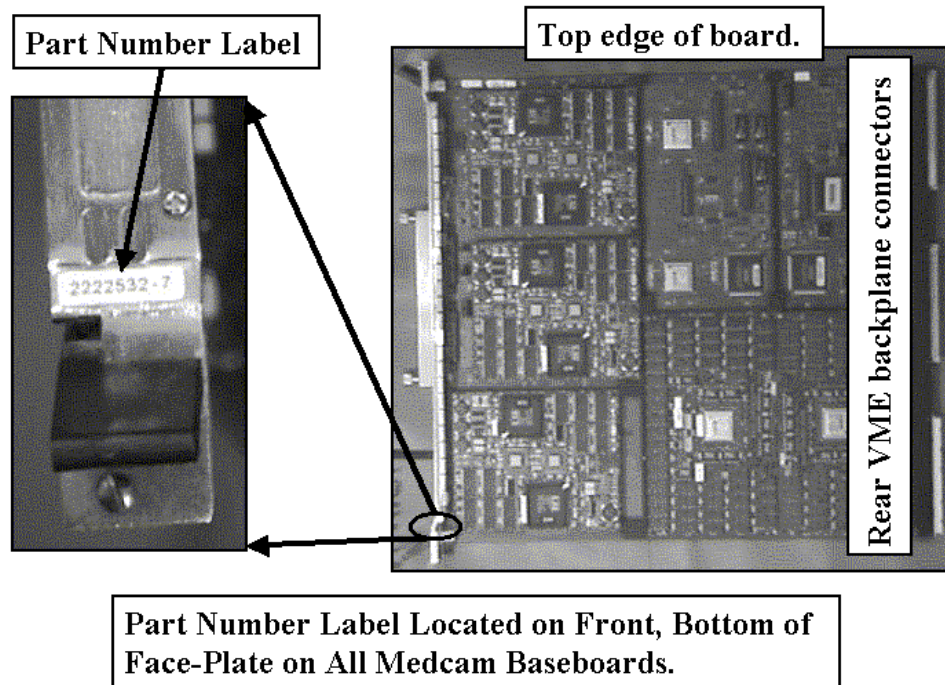


ILLUSTRATION 1
LOCATION OF MEDCAM BASEBOARD PART NUMBER

TABLE 1
MEDCAM COMPATIBILITY CHART

Medcam Baseboard Part Numbers	Medcam Baseboard Components								
	8 MB Single i860 CE Node (2101767-2)	128 MB MOD Memory (2101767-4)	1-Slot Foreplane Cover (2101767-5)	16 MB Dual i860 CE Node (2101767-8)	32 MB Single PPC Module (2222532-2)	64 MB Dual PPC Module (2222532-3)	Rev 0B Single 32MB PPC Module (2292662-2)	128 MB PMOD Memory (2292662-3)	Medcam Baseboard Terminator (2292662-6)
2101767-16, -18, -20, -22	X	X	X	X					
2222532-4, -5, -7, -8	X	X	X	X	X	X	X		
2292662-5, -7, -8, -9, -10			X			X	X	X	X
2292662-11*, -12, -13, -14			X			X	X	X	

(X – Signifies a compatible component.) *2292662-11 does NOT require jumpers.

2- REMOVAL OF FAILED MEDCAM AP BASEBOARD

1. Remove the System Cabinet front cover.
2. Turn off the main circuit breaker, CB1. (See Illustration 2-1.)

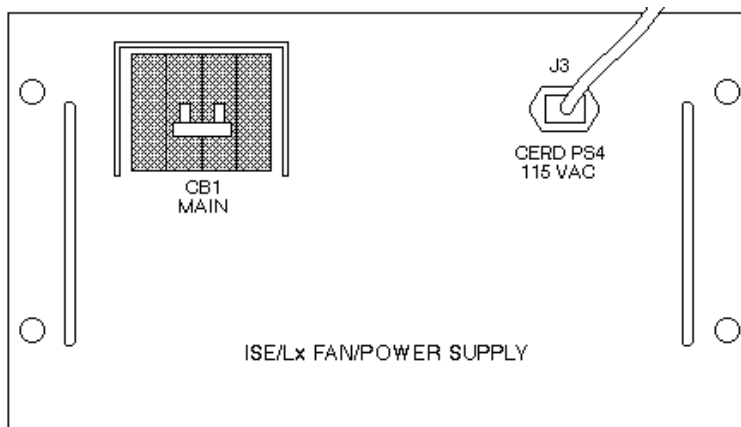


ILLUSTRATION 2-1
ISE POWER SUPPLY

3. Loosen the two front panel screws, and remove the AP assembly (Medcam baseboard) from the ISE/TPS Chassis. (See Illustration 2-2).

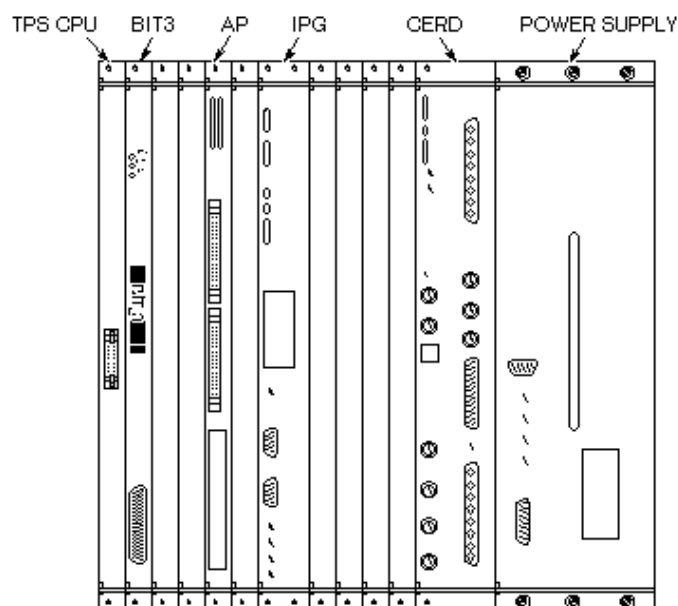


ILLUSTRATION 2-2
ISE/TPS CHASSIS

3- NEW MEDCAM AP BASEBOARD SET-UP

While the three boards may use different processor types, the board configuration is identical for the 2101767-16 and 2222532 type Medcam Baseboards and is described in *Section 3-1*. The configuration for the 2292662 type Medcam Baseboard is described in *Section 3-2*.

3-1 Configuration for 2101767-16 and 2222532 Type Medcam Baseboards

Illustrations 3-1, 3-2 and 3-3 show the standard configurations used in Signa Horizon Lx. Each illustration only shows the i860 processor module(s) attached. Be aware that the 2222532 type Medcam Baseboard can support **either** the i860 or PPC processor modules, but **not** mixed configurations.

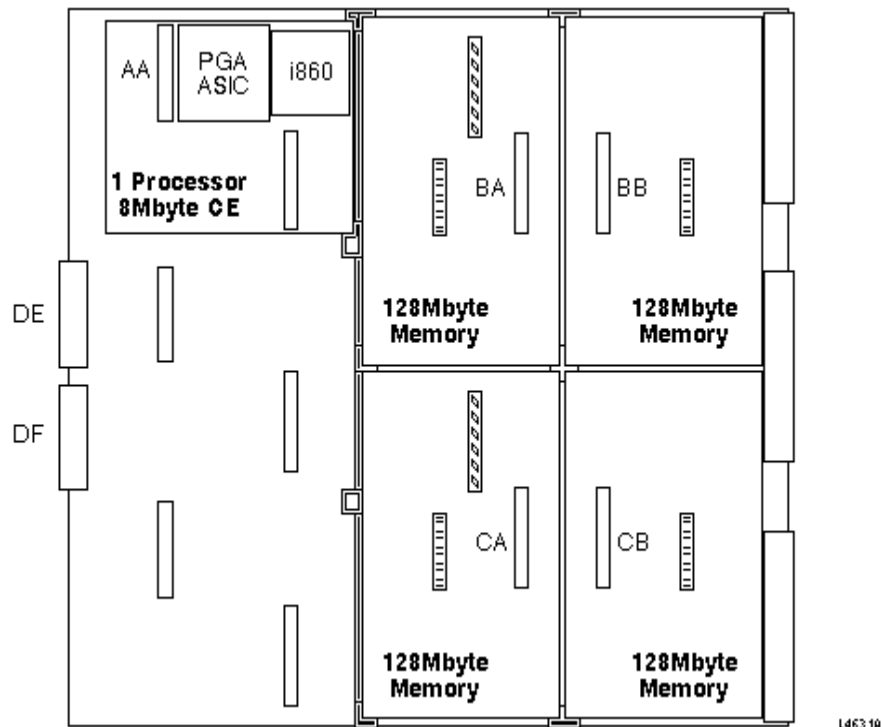


ILLUSTRATION 3-1
BASE AP (1 CPU; UP TO 512 MB MEMORY)

3-1 Configurations for 2101767-16 and 2222532 Type Medcam Baseboards (Continued)

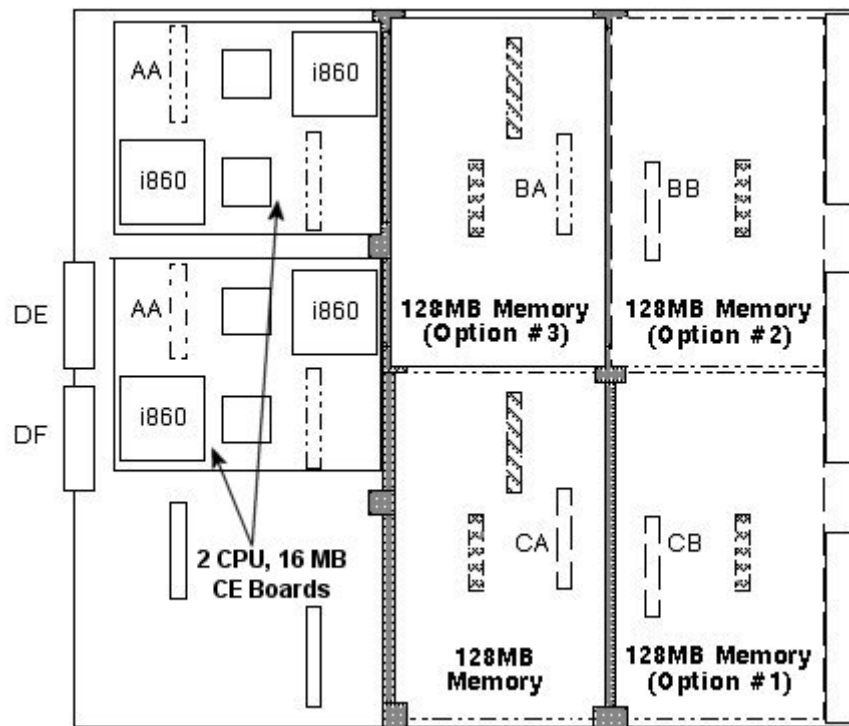


ILLUSTRATION 3-2
PERFORMANCE AP (4 CPU'S; UP TO 512 MB MEMORY)

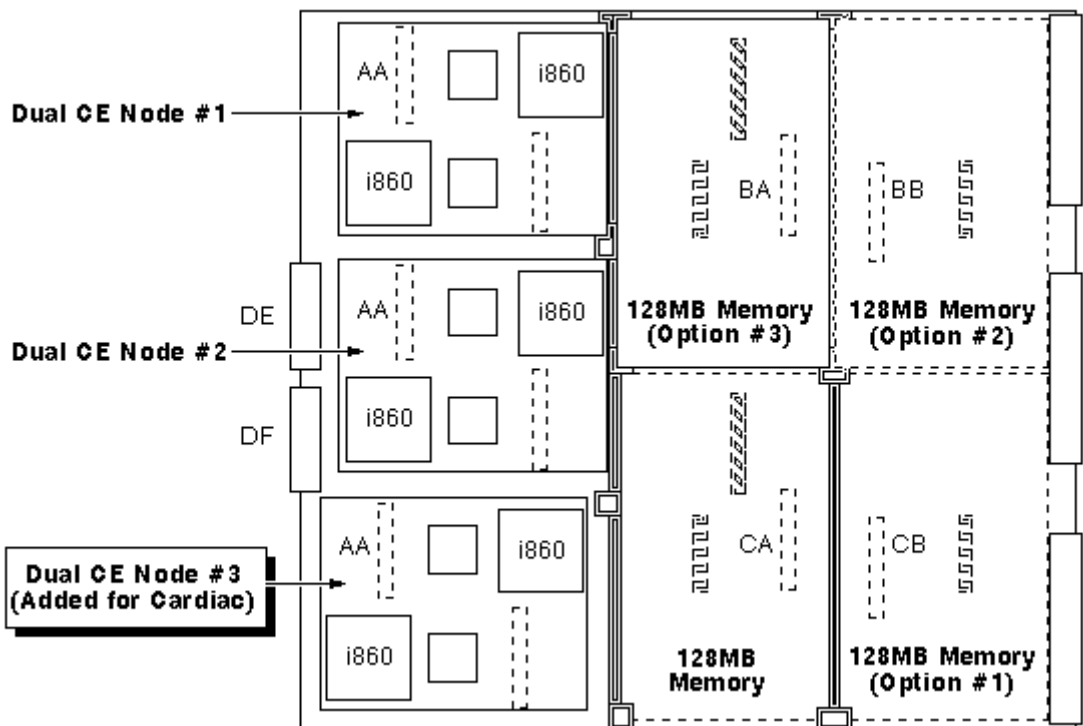


ILLUSTRATION 3-3
CARDIAC AP (6 CPU'S; UP TO 512 MB MEMORY ADDED FOR CARDIAC)

L463102

3-2 Configuration for 2292662 Type Medcam Baseboard

The configuration of the 2292662 type Medcam Baseboard is similar to its predecessors, with the exception that Baseboard 2292662-11 does NOT require jumpers. This board will also support up to six PPC CPUs (three dual PPC modules) and 512 MB of PMOD memory (four 128 MB PMOD memory boards).

3-3 Removing Modules from Any Type of Medcam Baseboard

1. Remove the 6 screws from the failed PPC Module or 8 screws from the failed 128 MB PMOD Memory Module.
2. Remove the failed module from the Medcam baseboard. If this is a 2292662 Medcam Baseboard and the module is not going to be replaced or the location it was removed from is going to be left temporarily open, refer to Illustration 3-4 and place terminators on the designated open PPC ports or memory nodes. (It may be necessary to order extra terminators, P/N 2292662-6, if extras are not available.)
3. Install the new module onto the Medcam baseboard and replace the screws. The orientation of the PPC module on the Medcam baseboard is not critical.

3-4 Adding/Upgrading Optional Modules on Any Type of Medcam Baseboard

1. For 2101767-16 or 2222532 type Medcam Baseboards: Refer to Illustrations 3-1, 3-2 and 3-3, and add the module(s) as shown.
2. For 2292662 type Medcam Baseboards: Refer to Illustration 3-4, and remove the terminators from the next open location(s) before installing the module(s). Save the terminators on site for possible future use.

3-5 Front Panel Switch Settings for All Medcam Baseboard Types

Switches found on the Medcam baseboard AP front panel are described in Table 3-1.

TABLE 3-1
AP BOARD FRONT PANEL SWITCH SETTINGS

Switch	Run Position	Description	Comments
Reset	Toggle	Forces a reset to the AP board when pushed.	Not often used.
S/W	0	Sets logic level for S/W.	Possible settings are 0 and 1.

3-6 Medcam Baseboard Jumper Settings

Refer to Illustration 3-4 for jumper locations and settings found on all the Medcam baseboard types used in the Signa Lx ISE/TPS chassis. Medcam baseboards should arrive with the jumpers properly configured. (Jumper configuration should not typically be necessary in the field.)

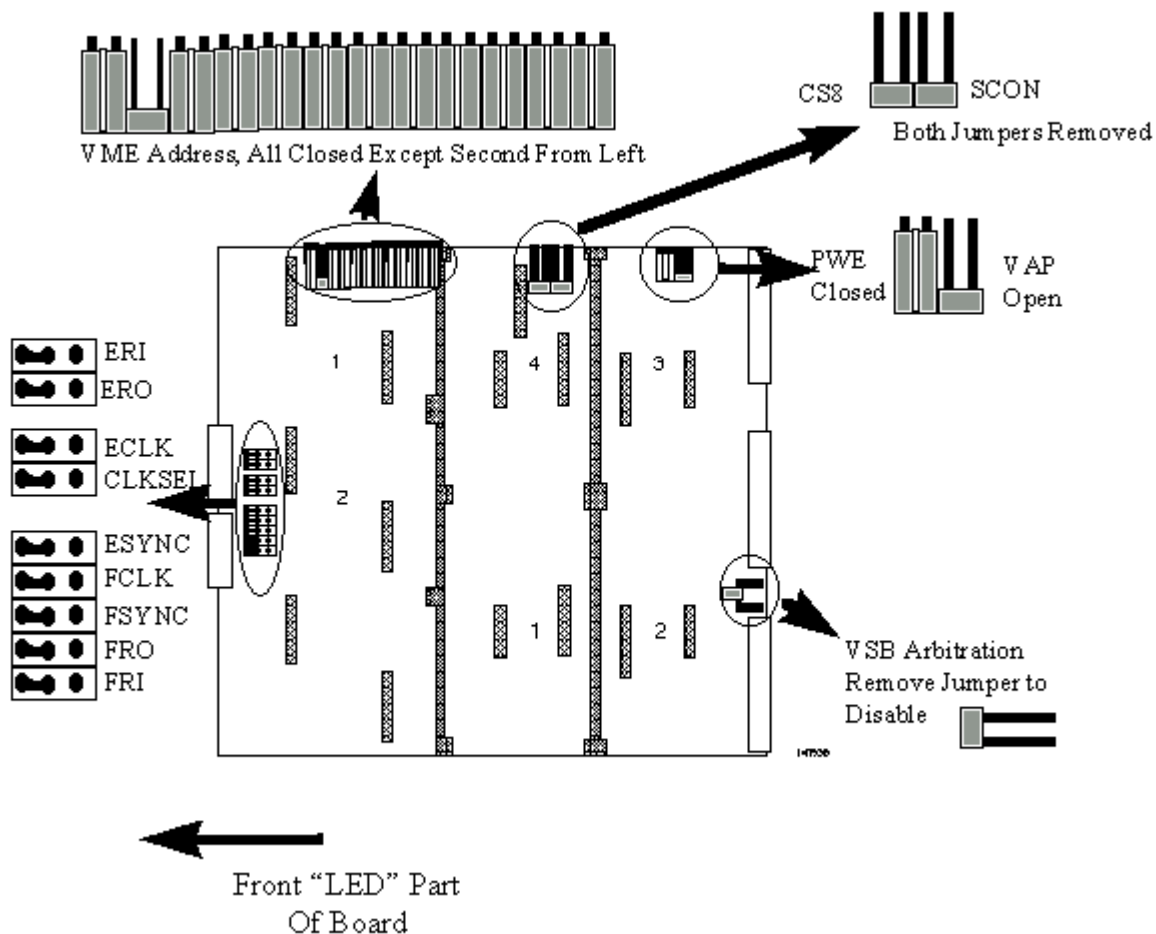


ILLUSTRATION 3-4
JUMPER LOCATIONS AND SETTINGS FOR ALL MEDCAM BASEBOARD TYPES

4- INSTALLATION OF REPLACEMENT BOARD

1. Install the Medcam Baseboard AP Assembly into the ISE/TPS Chassis and tighten the two front panel screws. (See Illustration 2-2.)
2. Turn on the main circuit breaker, CB1. (See Illustration 2-1.)
3. Install the System Cabinet front cover.
4. If Signa Horizon LX software is running, select **[Reset TPS]** from the Service Desktop.

4-1 Functional Checks Required

Perform one head or body scan.

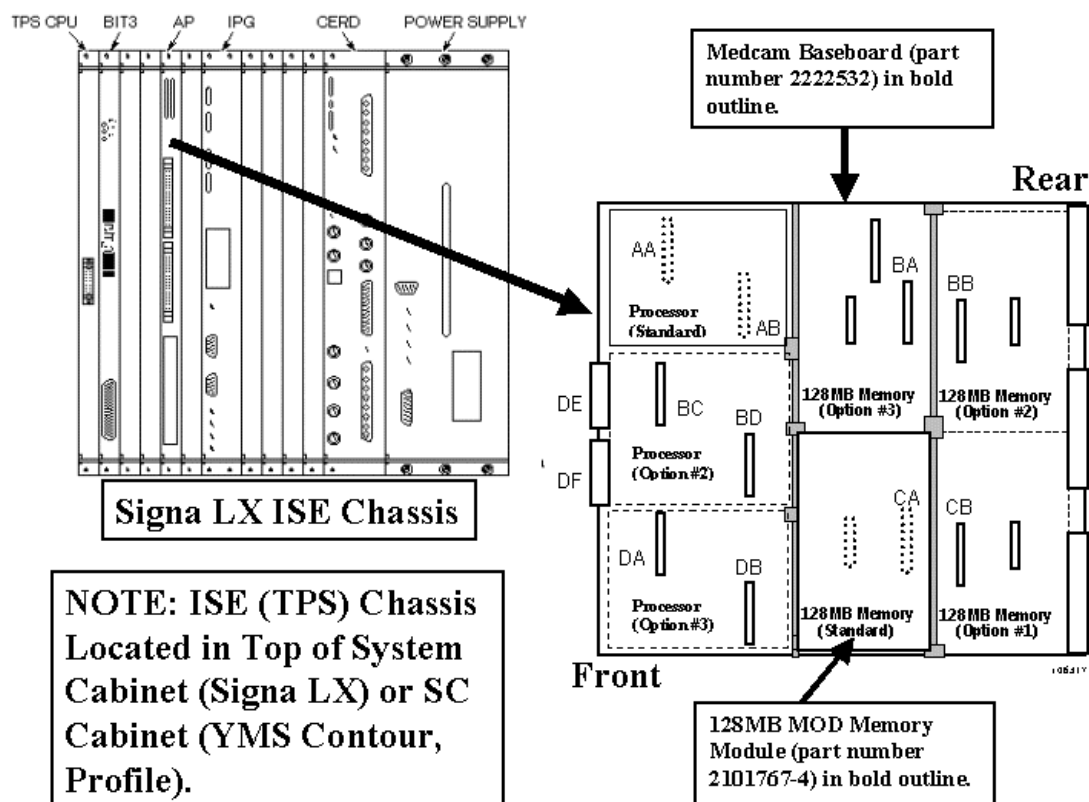
APPENDIX A – INTRODUCTION OF NEW MEDCAM MEMORY AND BASEBOARD (REPRINT OF SERVICE NOTE 67036)

Effectivity:

All Signa LX, CVi, NVi, 3T, SP and OpenSpeed MR scanners; all Profile 1, 2, or 3 and Contour 1, 2, or 3 MR scanners equipped with an SC Cabinet.

Problem:

The Medcam Baseboard, P/N 2222532, and the 128 MB MOD Memory Module, P/N 2101767-4 (components of a part also referred to as the Array Processor or AP) that reside in the ISE Chassis (sometimes referred to as the TPS) will no longer be produced by the vendor, Mercury Systems, resulting in an End of Life or EOL. (See Illustration A-1 below.)



LOCATION AND SIDE VIEW OF MEDCAM BASEBOARD SHOWING 128 MB MOD MEMORY MODULE
ILLUSTRATION A-1

Solution:

The vendor is producing a new Medcam Baseboard (P/N 2292662) and 128 MB PMOD Memory module (P/N 2292662-3) to replace the parts that have gone EOL. The new Medcam hardware is **not** backwards compatible with all the Medcam AP components currently in the field. The MR system **must** have A.S.P.2.0107 software or an equivalent version that has the drivers and diagnostics to support the new Medcam hardware. (See *Section 5, Determining the Software Version* and refer to Table A-1.)

TABLE A-1
SOFTWARE THAT SUPPORTS THE NEW MEDCAM HARDWARE

System	Software	Patch
Signa LX	A.S.P.2.0107 or equivalent	N/A
CVi, NVi	CNV3 or equivalent	N/A
3T	TBD*	N/A
SP	TBD*	N/A
OpenSpeed	M4 Release or equivalent	0044 Patch
Profile 1, 2, 3	No plans to update.	N/A
Contour 1, 2, 3	No plans to update.	N/A

* Some of the releases are yet to be determined.

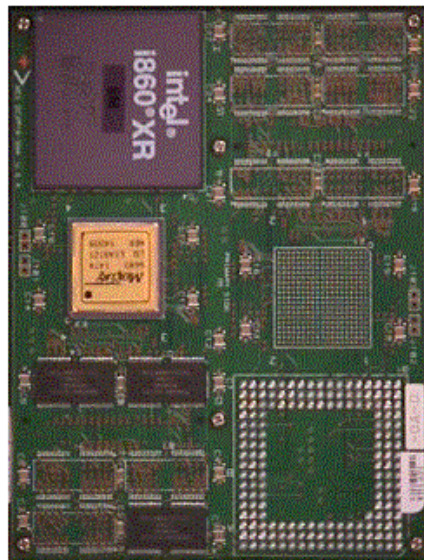
The new Medcam baseboard has special requirements. All unused processor ports and memory nodes on the new Medcam baseboard must be terminated. As a result, special terminators and a self-terminated Rev 0B Single PPC Processor Module (P/N 2292662-2) have been produced for the new Medcam baseboard. The only exception is that Baseboard 2292662-11 does **not** require jumpers. (See Table A-2 for an overview of the requirements for using the new Medcam hardware.)

TABLE A-2
OVERVIEW OF NEW MEDCAM AP FRUS

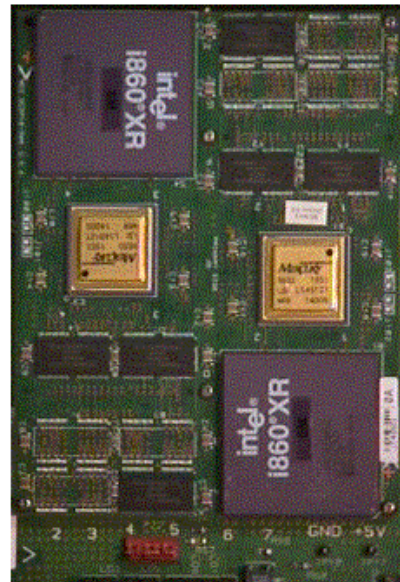
Part Number	FRU	Description	Requirements
2292662	1	Medcam Baseboard (See Section 1.)	Not compatible with the i860 processor. Must be used with the PPC Module. (See Illustrations A-2 and A-3.)
			All open memory nodes and PPC ports must be terminated to prevent TPS hangs. (See Illustration A-4.)
			Not compatible with older MOD 128MB memory (2101767-4).
			Reflex 20 configuration (single PPC not a current catalog option) must use the new Rev 0B Single 32 MB PPC Module with on-board termination, P/N 2292662-2.
			Requires A.S.P.2.0107 software or equivalent version. (See Table A-1.)
2292662-2	1	Rev 0B 32MB Single PPC Module with Termination (See Section 2.)	Required for Reflex 20 configuration (single PPC not a current catalog option) on the new Medcam Baseboard, P/N 2292662, to prevent TPS hang.
			Compatible with the old-style Medcam Baseboard, P/N 2222532.

Part Number	FRU	Description	Requirements
2292662-3	1	PMOD 128MB Memory Module (See Section 3.)	Not compatible with the i860 processor. Must be used with the PPC Module. (See Illustrations A-2 and A-3.)
			Not compatible with MDV9 Medcam Baseboard, P/N 2222532. (See Illustration A-7.)
			Not compatible with i860-only Medcam Baseboard, P/N 2101767-16. (See Illustration A-8.)
			Requires A.S.P.2.0107 software or equivalent version. (See Table A-1.)
			Mixing with old MOD 128 MB Memory, P/N 2101767-4, is not supported.
2292662-6	1	Medcam Baseboard Port Terminator (See Section 4.)	A terminator must be attached to each open memory node or PPC processor port on the new style Medcam Baseboard, P/N 2292662. Not required on the old board. The only exception is that Baseboard 2292662-11 does not require jumpers. (See Illustration A-5.)

* The old-style Medcam hardware FRUs will still be available and listed in Renewal Parts. Replace only with same FRU style.

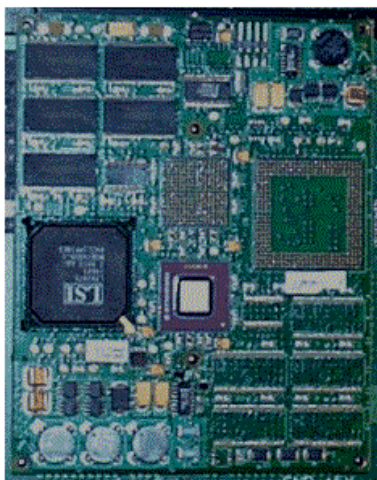


Single i860 Module

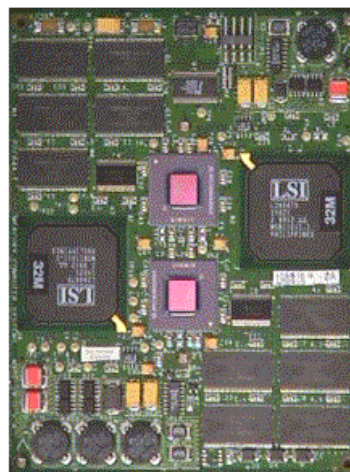


Dual i860 Module

ILLUSTRATION A-2
SINGLE I860 AND DUAL I860 MODULES



Rev 0B Single PPC Module w/term.



Dual PPC Module

ILLUSTRATION A-3

REV 0B SINGLE PPC MODULE WITH ONBOARD TERMINATION AND DUAL PPC MODULE

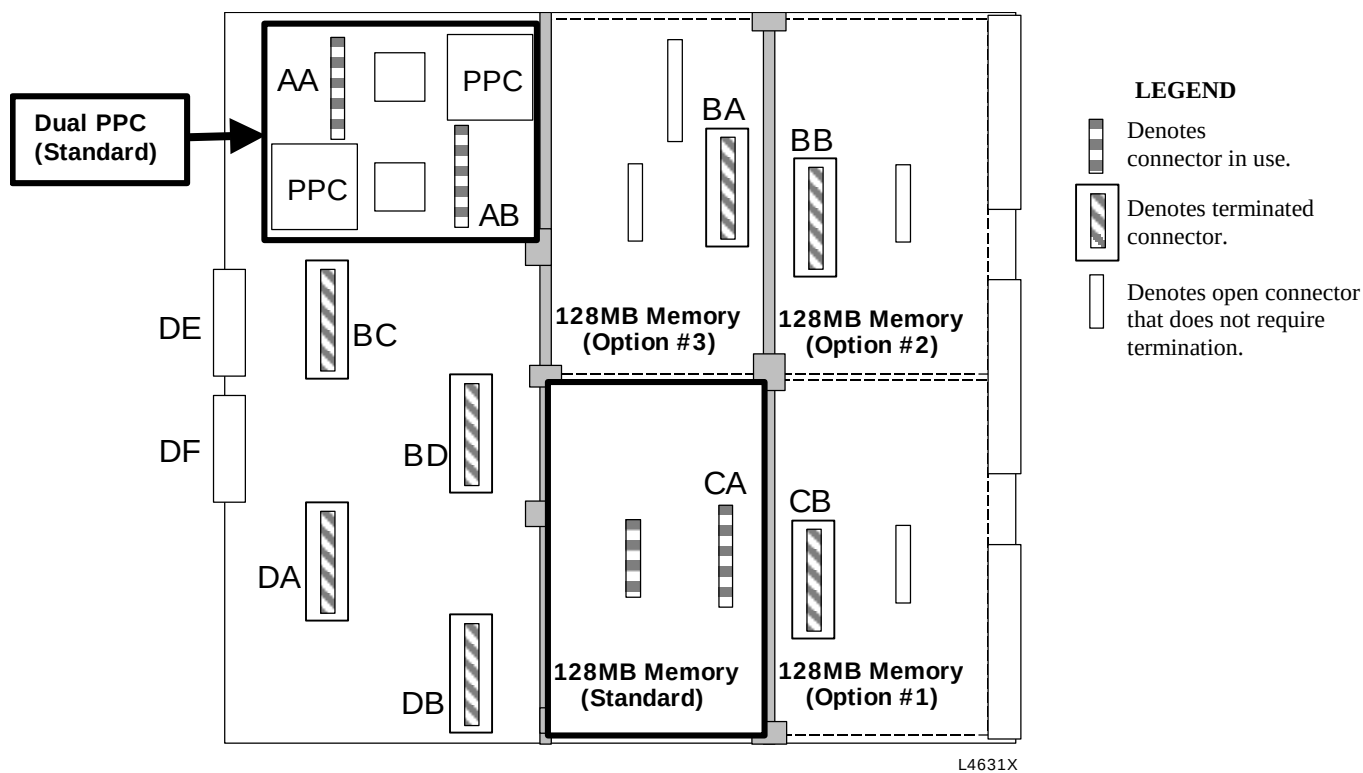


ILLUSTRATION A-4

TERMINATOR LOCATIONS FOR STANDARD NEW MEDCAM BASEBOARD (P/N 2292662)

NOTE: EXCEPTION IS BASEBOARD 2292662-11 DOES NOT REQUIRE JUMPERS.

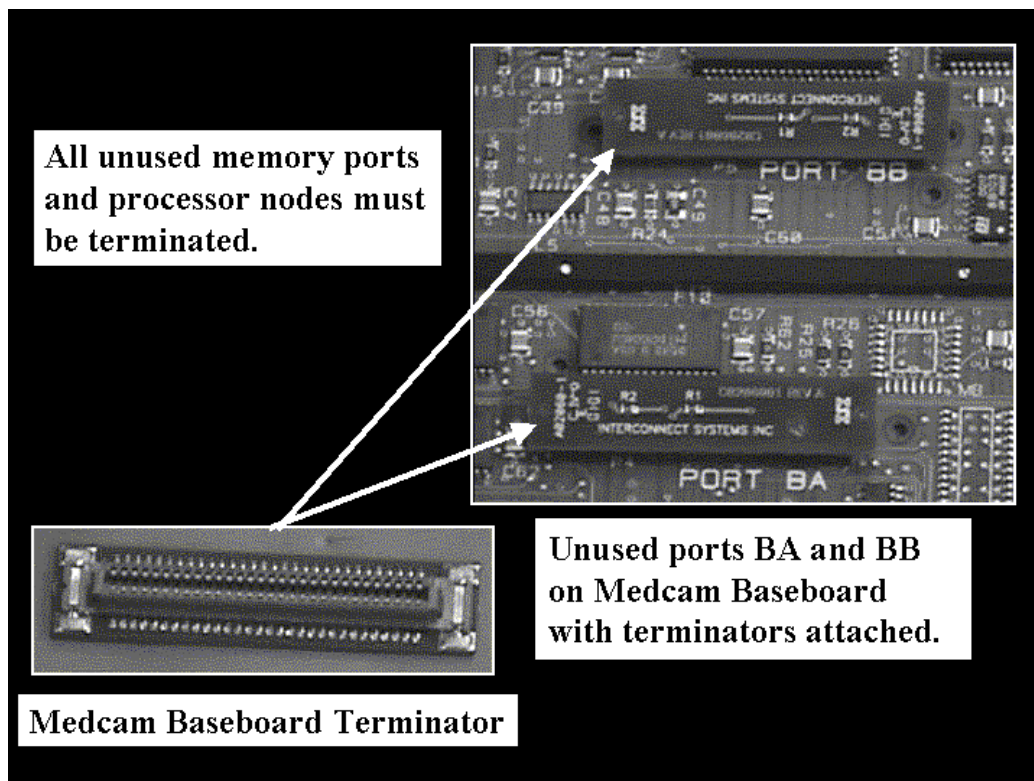


ILLUSTRATION A-5
MEDCAM BASEBOARD TERMINATORS

The new Medcam Baseboard (P/N 2292662) and the 128 MB PMOD Memory Module (P/N 2292662-4) will be used with the PPC processor, but not with any of the older Medcam components. When making repairs, be sure **not** to mix old and new Medcam baseboards and memory. The old-style Medcam Baseboard (P/N 2222532), which is being converted to a repairable part (T3), and the old-style MOD 128 MB memory modules (P/N 2101767-4, T3504P) will still be available as FRUs. Be careful to order an assembly with the correct PPC configuration. (See *Section 6, Identifying Medcam Components through Software* for instructions to determine processor type and quantity via software.

Forward production systems will begin shipping around fiscal week 10 with the new Medcam hardware. Except in certain unusual circumstances, the Medcam baseboard should arrive fully configured with all the necessary processors and memory preinstalled in the System Cabinet ISE Chassis. This was made possible when Marketing updated the recon/BAM catalog option offerings by phasing out catalogs M1000RL, M1000RM and M1000RN for Signa Lx and phasing in the following new standard configuration catalogs:

- M3000RA – Reflex 100 with 512MB PMOD memory (hex PPC configuration)
- M3000RB – Reflex 50 with 512MB PMOD memory (dual PPC configuration)
- M3000RC – Reflex 50 with 256MB PMOD memory (dual PPC configuration)

The path for upgrading the memory on the current installed-base, old-style Medcam will not change in the near future. Marketing will continue to supply the necessary MOD 128 MB memory upgrade packages for the old-style Medcam. In the distant future, however, a memory upgrade may require swapping out the entire Medcam assembly for the new one, and loading a new version of software containing the necessary drivers to support the new Medcam assembly.

There is currently no plan to upgrade the installed base to the new hardware since steps have been taken to ensure that the repair pool will be able to support the demand for the old style Medcam parts. Be careful to replace parts with **only** like parts. Do **not** attempt to mix old style and new style Medcam baseboards and memory as Medcam or memory errors may result.

Note

A processor(s) and/or on-board terminator will always be connected to ports AA and AB and a memory card, with the exception of certain Contour or Profile systems, which will always be connected to node CA. These ports and this node are already in use and do not need termination. Terminators will be added or removed from the other ports (BC, BD, DA and DB) and nodes (BA, BB and CB), as processors or memory modules are removed or added, respectively, from the new Medcam baseboard. The only exception is that Baseboard 2292662-11 does **not** require jumpers.

SECTION 1 – NEW MEDCAM BASEBOARD

The new Medcam baseboard looks similar to its predecessors. The i860-only Medcam Baseboard (P/N 2101767-16) and the PPC/i860 compatible MDV9 Medcam Baseboard (P/N 2222532), but there are some differences. The most notable difference is the appearance of the center-mounted chips on the boards. (See Illustrations A-6, A-7, and A-8.) The chips on the new Medcam board have a black finish, while the chips on the old Medcam boards have a gold finish.

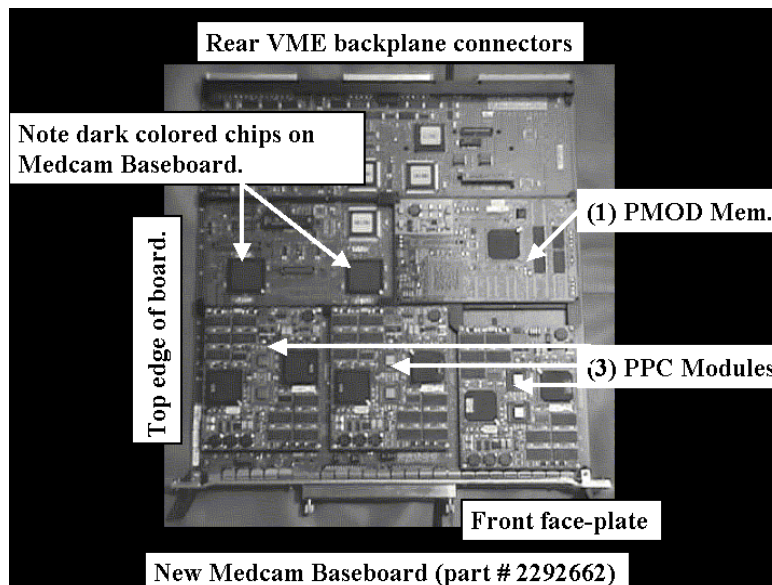


ILLUSTRATION A-6
NEW MEDCAM BASEBOARD

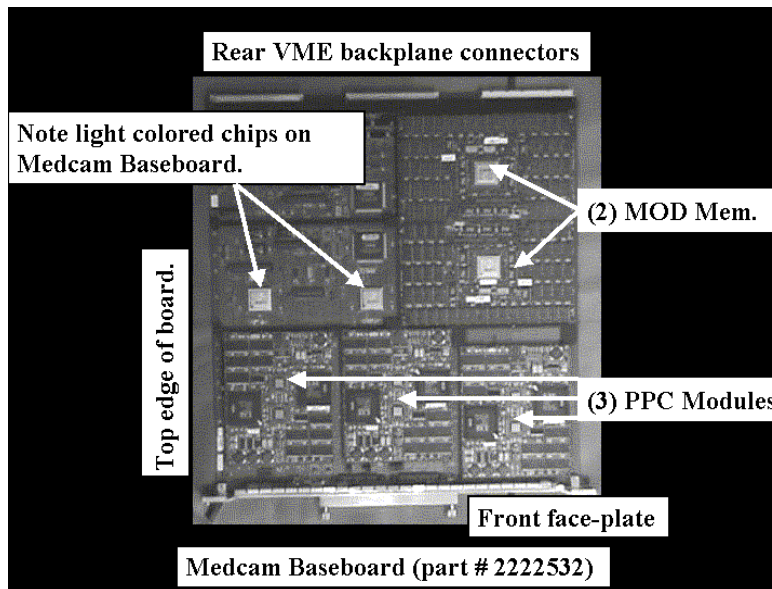


ILLUSTRATION A-7
MDV9 MEDCAM BASEBOARD

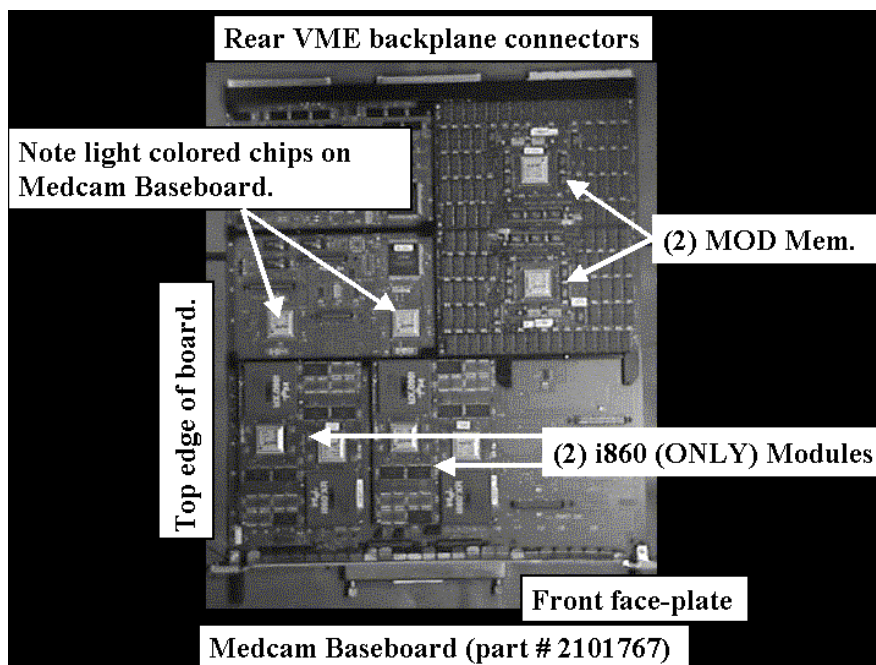


ILLUSTRATION A-8
ORIGINAL MEDCAM BASEBOARD (ONLY ACCEPTS I860 PROCESSORS)

All the boards should have a label affixed to the front, bottom of the board faceplate containing the part number. (See Illustration A-9.) The part number printed on the label of the new Medcam baseboard will read "2292662-X" where "X" may be:

- 4 (single PPC with 128 MB)
- 5 (dual PPC with 128 MB)
- 7 (hex PPC with 128 MB)
- 8 (dual PPC with 256 MB)
- 9 (dual PPC with 512MB) or
- 10 (hex PPC with 512MB)

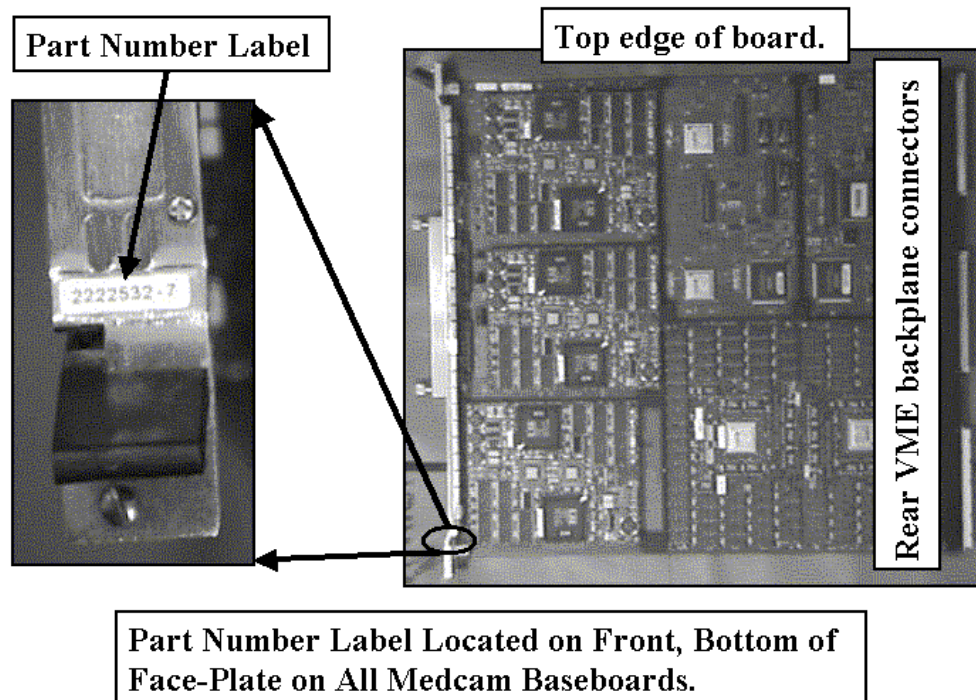


ILLUSTRATION A-9

PART NUMBER LABEL LOCATION ON ALL MEDCAM BASEBOARDS

All unused PPC ports and memory nodes on the new Medcam baseboard must be terminated with the supplied terminators. (The only exception is that Baseboard 2292662-11 does **not** require jumpers.) This refers to **only** the PPC ports and memory nodes on the Medcam baseboard. Depending on the number of processor modules installed on the Medcam baseboard, this might include ports BC, BD, DA and DB. (Ports AA and AB are not included, since a processor module should always be installed in this location.)

Depending on the number of memory modules installed on the Medcam baseboard, this might include nodes BA, BB and CB. Node CA is not included since, with the exception of certain Contour or Profile systems, a memory board should always be installed in this location. (See Illustration A-4 for a diagram showing the physical location of the memory nodes and processor ports on the Medcam baseboard.) Failure to terminate unused PPC ports and memory nodes will result in the TPS hangs.

SECTION 2 – REV 0B 32MB SINGLE PPC MODULE

The Rev 0B 32MB Single PPC module differs from its predecessor with on-board termination so that the second unused PPC port (AB or AA, depending on the orientation of the module on the Medcam baseboard) is terminated to prevent TPS hangs. It looks almost identical to the older 32 MB Single PPC Module (P/N 2222532-2), except it is labeled on the circuit board with the text “Rev 0B”. It is the functional equivalent of the older 32 MB Single PPC Module and can be used in its place. (See Illustration A-3.)

Note

Marketing is not currently planning to offer the Rev 0B 32 MB Single PPC Module (previously referred to as Reflex 20) as a standard configuration.

SECTION 3 – 128 MB PMOD MEMORY

The 128 MB PMOD Memory Module (P/N 2292662-3) looks much different from the old MOD 128 MB Memory Module (P/N 2101767-4). (See Illustration A-10.) It is not compatible with the i860 processor and requires A.S.P.2.0107 or equivalent software. (See Table A-1.)

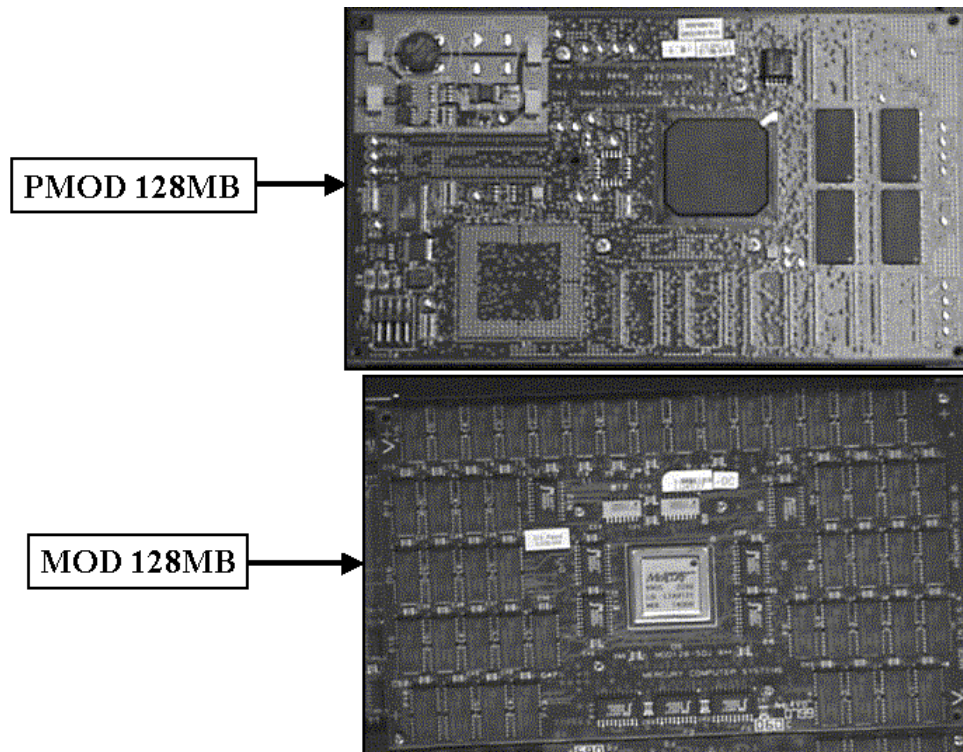


ILLUSTRATION A-10
COMPARISON OF PMOD AND MOD 128MB MEMORY MODULES

SECTION 4 – MEDCAM BASEBOARD TERMINATOR

All unused Medcam baseboard PPC ports and memory nodes must be terminated with the supplied terminators. (The only exception is that Baseboard 2292662-11 does **not** require jumpers.) This is not a requirement for all connectors on the Medcam baseboard, **only** the specified processor ports and memory nodes. A terminator consists of a connector attached to a small circuit board equipped with small resistive elements to provide the termination. (See Illustration A-4 for map of where the terminators will be placed, and Illustration A-5 for a picture of a terminator.)

SECTION 5 – DETERMINING THE SOFTWARE REVISION

Use the appropriate method described below to determine the revision of the software.

5-1 Determining Software Revision for a Signa Lx, NVi, CVi, 3T, or OpenSpeed System

1. Left mouse-click on the **C shell...** soft key located at the bottom of the Service Desktop Manager.
2. When a C-shell window opens at the bottom of the screen, place the mouse cursor inside the window.

3. After the prompt, type: **getver** and press **<Enter>**.

Output similar to the one below will display:

```
Hostname : <site_hostname>
Build number for MrpApps is ASP2.283B_M4_0107.a
Build number for PostSdC is ASP2.283B_M4_0107.a
Build number for cclass is ASP2.283B_M4_0107.a
Build number for driverSupport is ASP2.283B_M4_0107.a
Build number for insite is ASP2.283B_M4_0107.a
Build number for install is ASP2.283B_M4_0107.a
Build number for os_cd is 6.8.a
Build number for os_u_cd is 6u7.13.a
Build number for pc is ASP2.283B_M4_0107.a
MR Software release: A.S.P.2.0107a ←Note: Read the software revision here.
Any patches are listed prior to this line.
```

4. Close the C-shell window by holding down the **<Ctrl>** key and pressing the **D** key.

5-2 Determining Software Revision for a Signa Profile or Contour System

1. Left mouse-click on the **C shell...** soft key located at the bottom of the Service Desktop Manager.
2. When a C-shell window opens at the bottom of the screen, place the mouse cursor inside the window.
3. After the prompt, type: **config_status** and press **<Enter>**.
 - A window will display in the upper, left side of the screen with the message: "Gathering Data ... Please wait."
 - After approximately 30 seconds, the window will close and the "config_status" script will terminate. Approximately 5 seconds later, a larger window will display listing the system configuration.
 - The revision of the software and any included patches will be listed under the section labeled, **Software Revisions**.
4. Close the window by clicking on the **OK** soft key at the bottom of the window.

SECTION 6 – IDENTIFYING MEDCAM COMPONENTS THROUGH SOFTWARE

Since the Medcam baseboard part number is usually displayed on the front, bottom of the faceplate (shown in Illustration A-9), it is usually a simple matter to identify the Medcam baseboard without having to remove it from the ISE (also referred to as TPS) chassis. This is **not** the case for the modules attached to the Medcam baseboard and hidden from view in the ISE Chassis. It is possible, through software, to determine the Medcam baseboard processor type (i860 or PPC module), how many processors are in use, and the amount of memory installed on the Medcam baseboard. It is **not** possible to determine, through software, what type of memory (MOD or PMOD) is being used. The process for determining the Medcam status by querying software differs based on whether the system is Signa Lx or Contour/Profile. Both processes are described below.

6-1 Querying Software for Signa LX Medcam Component Status

1. Left mouse-click on the **C shell...** soft key located at the bottom of the Service Desktop Manager.
2. When a C-shell window opens at the bottom of the screen, place the mouse cursor inside the window.
3. After the prompt, type: **more /usr/g/bin/medcam.cfg** and press **<Enter>**.

The resulting display will be similar to the one below with the exception that there will be only one outcome listed for each line of the display:

Number Recon Processors: **6** (Hex) or **4** (Quad) or **2** (Dual) or **1** (Single)
Amount of Recon Memory: **128** or **256** or **384** or **512** Mbytes
Processor Type: **860** or **PPC**

4. Close the C-shell window by holding down the **<Ctrl>** key and pressing the **D** key.

6-2 Querying Software for Contour/Profile Medcam Component Status

1. Left mouse-click on the **C shell...** soft key located at the bottom of the Service Desktop Manager.
2. When a C-shell window opens at the bottom of the screen, place the mouse cursor inside the window.
3. After the prompt, type: **config_status** and press **<Enter>**.
 - A window will display in the upper left side of the screen containing this message: "Gathering Data . . . Please wait."
 - After approximately 30 seconds, the window will close and the "config_status" script will terminate. Approximately 5 seconds later, a larger window will display listing the system configuration.
4. Scroll to the bottom of this larger window, and check the memory size and processor type under the section labeled, **Hardware Configuration**. (See Table A-2 for a summary of the various Medcam configurations.)
5. Close the C-shell window by holding down the **<Ctrl>** key and pressing the **D** key.

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
0	May 6,1998	R. Hawthorne	Initial update from Toolbook to Word. Also added AP board jumper map.
1	April 18, 2001	D. Thome	Added Compatibility Table 1 and Illustration 2-4 for new Medcam Baseboard. Added comments concerning new Medcam Baseboard. Included Service Note 67036 in Appendix A.
2	Nov. 6, 2008	L. Hannon	Revised Table 1 for Medcam Baseboard 2292662, a new part that no longer needs terminators per PQR 13165487.
3	Oct. 15, 2009	L. Hannon	Added a Note to Table 1 and Section 3-2 Configuration for 2292662 Type Medcam Baseboard for 2292662-11 per SPR SVCge48089.